INCH-POUND

MIL-B-45993A
9 June 1989
SUPERSEDING
MIL-B-45993(AL)
21 February 1984

MILITARY SPECIFICATION

BEAD BREAKER, PNEUMATIC TIRE, PORTABLE AIR OR AIR OVER HYDRAULIC OPERATED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an air or air over hydraulic operated portable hand held tire bead breaker for the wheel/tire assemblies of military vehicles and all commercial vehicles with the size range specified herein.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issue of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-601	- Boxes, Wood, Cleated-Plywood
PPP-B-636	- Boxes, Shipping, Fiberboard
PPP-T-60	- Tape, Packaging, Waterproof

MILITARY

MIL-P-116	- Preservation, Methods of
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-C-4109	- Coupling Halves, Low Pressure, Air Hose, Quick-Disconnect
MIL-C-5501	- Caps and plugs, Protective, Dust and Moisture Seal
MIL-H-6083	- Hydraulic Fluid, Petroleum Base, For Preservation and

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-EST-S, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Operation
MIL-H-13531 - Hose, Rubber and Hose Assembly, Rubber (Hydraulic Flexible)
MIL-F-17111 - Fluid, Power Transmission

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-1190 - Minimum Guidelines for Level C Preservation, Packing and Marking

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publication. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issue of the documents cited in the solicitation.

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE J517 - Hydraulic Hose

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., Two Pennsylvania Plaza, New York, NY 10001.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2.2.1, 6.2, and 6.3).

3.2 Design. The bead breaker shall be designed to separate a tire bead from its wheel rim. It shall be capable of being operated by one person. It shall break beads of all tube and tubeless type tires from flat-base and drop center automotive rims within a size range of 7.00 by 15 through 14.00 by 24 and 15.00 by 19.5. The

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bead breaker shall consist of a hand rim clamping and bead breaking assembly operated by air, or a hand air over hydraulic pump assembly. Additional tools, adapters, or devices shall not be required for fastening or positioning on wheel to perform the bead breaking operation. The bead breaker shall perform all requirements herein without malfunction, breakage, damage (see 6.6) or deformation of any component, leakage, or damage to the tire or rim. Total weight of the bead breaker, with pump and hose assembly, shall not exceed 50 lbs.

3.2.1 Maintainability. Design and configuration of the bead breaker and the adjustment features shall assure that maintenance of operational capability and accuracy will be reduced to the simplest procedures and require a minimum of skills and common maintenance equipment. Instruction for disassembly, repair, assembly, and adjustment will be clear, concise, and definitive in application.

3.3 Safety and health requirements. Covers or other safety devices shall be provided for all parts of the bead breaker presenting safety hazards which can be guarded without substantial interference to operation of the bead breaker. All guards shall be removable to provide easy access to guarded parts. Operation of the bead breaker shall pose no hazard to the operating personnel.

3.4 Construction. The bead breaker shall be so constructed to be complete so that when connected to an air supply it can be used for the operations specified herein.

3.4.1 Raw materials. The manufacturer may use reclaimed materials for fabricating new parts. Reclaimed materials shall be reprocessed, remanufactured, or recycled in a manner that restores them to the same chemical composition and physical properties as the raw materials originally selected for use. Use of reclaimed parts as is or rebuilt from scrap or other used equipment shall not be permitted.

3.4.2 Interchangeability. All parts of the bead breaker shall be manufactured to definite standards and tolerances that will provide for the interchangeability of respective parts between end items of the same model. All replacement parts shall be interchangeable with their respective original parts, and fit/function with mating components.

3.4.3 Castings and forgings. Finished castings and forgings shall be free of fins, scale, inclusions, cold shuts, voids, cracks, thermal ruptures, laps, folds, mismatching, etc. No processes such as welding, peening, plugging, soldering, or pasting shall be used for repairing, reworking, or improving cosmetic appearance.

3.4.4 Welding, brazing, or soldering. All welding, brazing, and soldering shall be free of defects and of a quality which shall sustain all requirements of the welded, brazed, or soldered parts. Defects shall include but are not limited to such things as bubbles, cracks, gouges, holes, insufficient material used, etc. All flux or other such materials used during these operations shall be thoroughly removed (cleaned), from the part(s) immediately upon completion of the operation. These operations shall not be employed as repair measures for defective parts.

3.4.5 Fastening devices. Fastening devices and methods shall be chosen to serve the need while providing necessary adjustability for service, maintenance or repair. Fasteners shall be installed to prevent loss of tightness and shall not loosen in service.

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3.4.6 Threads. All machined threads shall conform to FED-STD-H28, and the applicable detail standard referenced therein.

3.5 Components and product characteristics.

3.5.1 Rim clamping and bead breaker assembly. The bead breaker assembly shall consist of a mechanism to perform the action of clamping the bead breaker assembly onto the rim and simultaneously applying a progressive clamping and wedging force between the tire bead and the rim flange. The component(s) which effect breaking of the tire bead from the rim shall maintain metal to metal contact with the rim flange as it follows the rim contour down the rim bead seat. No repositioning of the bead breaker assembly shall be required once the rim clamping, wedging action has begun. The component(s) which effect breaking of the tire bead from the rim, shall not slip from the tire bead to the tire sidewall during the bead breaking operation. While the progressive clamping is sustained (or increased), the bead breaker component(s) shall automatically return to their retracted position. The assembly may be repositioned and the bead breaking operation re-performed around the tire and rim assembly for total separation of the tire bead from the rim. The assembly shall have a rim width clamping range from .75 inch to 3.5 inches and a pressure foot travel of not less than 3.5 inches. As a minimum, the bead breaker shall be capable of breaking the tire beads for the rims/wheels specified in 3.2 and the following:

a. The wheels and tires from the Army's Commercial Utility Cargo Vehicle (CUCV), which is General Motors Corp., Chevrolet Blazer pick-up truck, which are identified as:

(1) Wheel - Fifteen inch x 8.0 inch General Motors Part Number 14063307, 6 hole lug pattern, 5.5 inch diameter bolt circle, with 0.66 inch positive offset (inset). Tire - 10R15LT, on/off road, 6 ply (Load Range C).

(2) Wheel - Sixteen inch x 6.5 inch General Motors Part Number 14035374, 8 hole lug pattern, 6.5 inch diameter bolt circle, with 0.67 inch positive offset (inset). Tire - LT235/85-R-16, on/off road, 10 ply (Load Range E).

b. The wheels and tires from the M880 Recovery Vehicle, which is Chrysler Corp., Dodge Power Wagon, which are identified as:

(1) Tire size L78-15/9.00 x 15, wheel mounted.

(2) Wheel 6.75 x 16.5 inch, Chrysler Part Number 4026506, with a mounted tire size of 9.50R x 16.5D, 8 ply. For information, the wheel has an 8 hole lug bolt pattern, a 6.5 inch diameter bolt circle, and a 4.78 inch diameter centering hole.

c. Wheel - Commercial wheel with 15 inch diameter and from 6 to 7 inch width bead to bead. Tire - Commercial tire, 9.00 x 15, 2 ply (4 ply rated).

d. The wheels and tires from the M198 Howitzer, identified as Goodyear Supersingle KK95952B, 16.5 by 19.5, 16 ply rated, tubeless mounted on the wheel that comes with the vehicle.

e. The wheels and tires from the M747 Trailer, Identified as Firestone Part No. 2F-29G-22-6-95 mounted on the wheel that comes with the trailer.

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3.5.2 Air actuated bead breaker. The air actuated bead breaker shall comply with the stated performance requirements specified in 3.5.1. It shall be actuated by an air (only) supply from 95 to 105 pounds per square inch gage.

3.5.3 Air over hydraulic bead breaker.

3.5.3.1 Air over hydraulic pump assembly. The bead breaker shall be actuated by an air driven hydraulic pump activated by a foot treadle. The air over hydraulic pump assembly shall be capable of developing sufficient pressure for operation of the rim clamping and bead breaker assembly as specified in 3.5.1. The air supply pressure required for operation of pump assembly shall be within the range of 95 to 105 pounds per square inch. The pump shall be provided with a built-in safety overload valve and a control valve to allow instant release of pumping action. The reservoir shall have ample capacity for total displacement of the bead breaker lug 15 percent or more. The hydraulic reservoir of each bead breaker supplied shall be filled to the manufacturer's operating level with hydraulic fluid. The manufacturer shall indicate either MIL-F-17111 and MIL-H 6083 as the applicable fluid to be used for refilling the bead breaker. If the bead breaker uses hydraulic fluid meeting the requirements of MIL-F-17111, the following message shall be attached in a conspicuous location near the filler plug of the reservoir:

CAUTION
USE NO OTHER FLUID THAN
MIL-F-17111

If the bead breaker uses hydraulic fluid meeting the requirements of MIL-H-6083, the following message shall be attached in a conspicuous location near the filler plug of the reservoir:

CAUTION
USE NO OTHER FLUID THAN
MIL-H-6083

3.5.3.2 Hydraulic hose assembly. The bead breaker hose assembly between the pump and bead breaker assemblies shall be not less than 72 inches long. Hose shall comply with requirements of Standard SAE J517-100R2A or MIL-H-13531, Type II, Class B. Each end of the hose shall be furnished with a wire spring guard. The burst pressure of the hose shall be rated at 4 times higher than the maximum possible working pressure.

3.5.3.3 Hydraulic quick disconnect coupling. There shall be a quick disconnect coupling between the air over hydraulic pump assembly hose and the bead breaker assembly. It shall be of the double end shut-off type with 360 degree swivel action and dust caps.

3.5.4 Air line quick disconnect couplings. There shall be two quick disconnect couplings provided. They shall be the Type I, Style 2 and the Type II, Style 1 of MIL-C-4109. The Type II, Style 1 shall be attached to the bead breaker air line port.

3.6 Painting and protective finish. Painting and protective finishing shall be in accordance with the best practice of commercial industry. The protective finish shall be a paint coating of at least 2.0 mils (0.002 inch) thickness. All surfaces to be painted shall, immediately prior to painting, be cleaned, free of all foreign matter, and dried. Paint shall adhere to the surface of the end item, when fiberglass tape is removed from two intersecting knife cuts, at least two inches

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long, through to the base metal. The surface hardness of the paint shall be between the industrial standards of H to HB pencil lead. Where no protective finish is utilized, the natural finish of the material or the finish obtained from heat treatment is permissible provided the surfaces are free of scale and have a protective coating to inhibit corrosion.

3.7 Identification marking. The bead breaker shall be marked for identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2), shall include the National Stock Number.

3.8 Workmanship. The bead breaker, including all parts and accessories, shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, scale, and sharp edges; marking of parts and assemblies; thoroughness of welding, brazing, painting, and riveting; alignment of parts and tightness of assembly screws and bolts.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (see 4.2.1).

b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. First article inspection shall be applied to the preproduction model or initial production item and to the first item produced subsequent to a significant design change in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.3 and the tests in 4.4, 4.5 and 4.6 Failure of the first article to pass examination inspection or the tests shall be cause for rejection of the unit.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be applied to the production items being offered for acceptance under the contract. Quality conformance inspection shall consist of (a) through (d) as follows. Failure of an item to pass an examination or test shall be cause for rejection of the unit.

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a. Examination (see 4.3).
b. Performance test (see 4.4).
c. Paint test (see 4.6).
d. Packaging inspection (see 4.7).

4.3 Examination. Visually and manually examine each bead breaker to determine conformance with 3.2 through 3.8. Visual examinations shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from the identified defects. Manual examinations shall include the operation of moveable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the bead breaker and its components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection. In addition, failure of the contractor to provide objective evidence that all parts are manufactured to definite standards. clearances, and tolerances so all replacement parts will not degrade the form, fit, and function of the end item (see 3.4.2) shall be cause for rejection.

4.4 Performance test. Each bead breaker under test shall be operated in accordance with the manufacturer's instructions to break the beads on both sides of tires of the sizes 14.0 x 24, 20 ply; 7.00 x 16, 6 ply; 6.50 x 16, 10 ply; 9.00 x 15, 2 ply (4 ply rated); and 9.00 x 15, 12 ply. Each test bead breaker shall be used on at least three tires and wheels. All tires used in testing shall be obtained from vehicles that have been in service in order that conditions will be encountered similar to those found under actual field conditions. Air supply pressure shall be from 95 to 105 pounds per square inch gage. Inability of the bead breaker to break the tire bead free from the wheel rim or evidence of any hydraulic fluid leakage or bead breaker malfunction or damage during operation shall be cause for rejection. The tire shall be removed and examined for damage to the beads and sidewalls, and the rim flange and bead seat band shall be examined for deformation and gouging. The tire shall be mounted and inflated in accordance with the tire manufacturer's instructions and examined for leaks due to damage or deformation caused by the bead breaker (see 3.2).

4.5 Rim clamping and bead breaker assembly test. The bead breaker shall be tested in accordance with 4.4 to determine compliance with 3.5.1, including capability throughout the range of specified tire and wheel sizes, including each tire size listed in 3.2. Inability of the bead breaker to break the bead free from the wheel rim and to perform the function and procedure described in 3.5.1 shall be cause for rejection.

4.6 Paint test. The hardness, adhesion and thickness of the paint used on the bead breaker shall be checked as follows to determine conformance with 3.6:

(a) The graphite point of a sharpened HB lead pencil shall be pressed at a right angle into a painted area of the bead breaker until the graphite crumbles. Failure of the surface to withstand the test, as evidenced by a pitting of of the paint, shall be cause for rejection.

(b) A test sample of the same metal, painted with the same process and at the same time as the bead breaker, shall have a sharpened H lead pencil pressed into the paint as in (a). Failure to penetrate the surface of the paint down to the base metal (paint too hard), shall be cause for rejection.

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(c) Two, two-inch intersecting lines shall be cut through the paint to the base metal of the test sample. A strip of 1/2 inch wide fiberglass tape shall be firmly pressed over the intersection and then pulled from the surface. Adhesion of any paint to the tape shall be cause for rejection.

(d) Using a paint thickness measuring instrument, or by masking a portion of the test sample during painting, measure the thickness of the paint. Failure of the paint to be at least 0.002 mil thick shall be cause for rejection.

4.7 Packaging inspection. Inspection to determine compliance with cleaning, preservation, packaging, packing, and marking requirements of the applicable packaging documentation, for the level or degree of protection designated in the contract, shall be as specified in section 5.

5 PACKAGING

5.1 Preservation. Preservation shall be Level A or C, as specified (see 6.2).

5.1.1 Level A preservation.

5.1.1.1 Cleaning and drying. Components of the bead breaker shall be cleaned and dried in accordance with the applicable procedures of MIL-P-116.

5.1.1.2 Preservation application. All exposed, critical ferrous metal surfaces of the bead breaker shall be coated with type P-2 preservative. Non-critical, unpainted, ferrous metal surfaces shall be coated with a preservative compound conforming to type P-1. When applicable, the interior surface of the air motor shall be preserved with type P-10, grade 30 preservative oil. Preservatives specified and the procedures for application shall conform to MIL-P-116.

5.1.1.3 Unit protection. The method of unit protection for the bead breaker shall conform to method I to MIL-P-116. Surfaces coated with type P-2 preservative oil shall be covered with barrier material conforming to MIL-B-121, grade A. The material shall be secured in place with tape conforming to PPP-T-60, type optional. Plastic cap-plugs conforming to MIL-C-5501 shall be placed over the metal threads of the air and hydraulic fittings. The bead breaker assembly along with the hydraulic hose (see 5.1.1.4) and the technical data (see 5.1.1.5) shall be packaged in a close-fitting fiberboard box conforming to PPP-B-636, grade V3c. Sufficient blocking, as specified in MIL-P-116, shall be applied to immobilize the contents of the box. The box closure shall be in accordance with the appendix to the container specification.

5.1.1.3.1 Hose assembly. The hose assembly shall be coiled to a minimum safe diameter and secured with cotton tape or twine to prevent uncoiling. The hose shall be packaged in accordance with method IC-3 of MIL-P-116.

5.1.1.3.2 Technical data. Publications issued with the bead breaker shall be packaged in accordance with method IC-1 of MIL-P-116.

5.1.2 Level C. The bead breaker shall be cleaned, preserved, and packaged in accordance with MIL-STD-1190.

5.2 Packing. Packing shall be level A or B, or C, as specified (see 6.2).

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5.2.1 Level A packing. Four each bead breakers shall be packed in a snug-fitting shipping container conforming to PPP-B-601, overseas type. The box closure shall be in accordance with the appendix to the container specifications.

5.2.2 Level B packing. Packing requirements shall be the same as for level A (see 5.2.1), except the shipping container shall conform to PPP-B-601, domestic type.

5.2.3 Level C packing. Level C packing of the bead breaker shall conform to MIL-STD-1190.

5.3 Marking. Marking shall be Military in accordance with MIL-STD-129 or MIL-STD-1190 as specified in the contract (see 6.2).

6 NOTES

(This section contains information of a general explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The bead breaker covered by this specification is a portable bead breaker for use in field shops to facilitate tire removal from military vehicles and commercial light and heavy trucks. It is intended for use to break beads of tube and tubeless type tires from flat-base and drop center passenger car and truck/bus rims within the size range of 7.00 by 15 through 14.00 by 24.

6.2 Acquisition requirements. Procurement documents should specify the following:

a. Title, number, and data of this purchase description.
b. First article, when required (see 3.1).
c. Nameplate, if different (see 3.7).
d. First article inspection, if different (see 4.2.1).
e. Degree of preservation and packaging, and degree of packing required (see 5.1 and 5.2).
f. Marking, as specified (see 5.3).
g. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2).

6.3 First article. When a first article inspection is required, the attachments should be a first article sample, a first production item, a sample selected from the first run of production items or it may be a standard production item from the contractor's current inventory as specified in 4.2.2.1. The first article should consist of a specific number of units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement of samples for first article inspection to those biders offering products which has been previously acquired or tested by the Government, and that bidders offering such products who wish to reply on such production or test, must furnish evidence with the bid that prior Government is presently appropriate for the pending contract.

6.4 Data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1423 incorporated into the contract.

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6.5 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.6 Definitions. The definition used in this document is defined as follows: Damage to the bead and sidewalls - Damage to the bead and sidewalls is defined as internal and external structure damage and gouging, tearing, cutting or slitting of the tire structure of carcass, and damage to the rim flange and bead seat band which causes deformation of the metal beyond its initial manufactured state. Superficial scratches or surface dents that do not change or modify the initial configuration of the rim/wheel assembly is not considered to be deforming in nature.

6.7 Subject term (key word) listing.

Bead Breaker
Interchangeability
Threads

6.8 Sampling. Sampling for quality conformance inspection should be performed as follows. Lots should be accepted with zero defects and rejected if one or more defects are found.

Inspection or test	Lot or batch size	Number randomly selected for inspection
Examination (see 4.3)	100 percent	---
Performance (see 4.4)	1 - 150	8
Paint test (see 4.6)	1 - 150	5
Packaging inspection (see 4.7)	1 - 150	4.0

The lot size should not exceed the maximum sizes indicated above.

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Air Force - 99

Preparing activity:
Army - AL

Review activities:
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DOCUMENT IDENTIFIER AND TITLE MIL-B-45993, Revision A, Bead Breaker, Pneumatic Tire, Portable Air or Air Over Hydraulic Operated

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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